

→ Density-based algorithm that discovers clusters of arbitrary shape and automatically flags outliers as noise. Perfect for anomaly detection in manufacturing process data where defects are rare but critical.

Silhouette Score : 0.9829

N Noise Points : 0

[illegible]

After training and comparing all recommended models, KMeans achieved the best performance. It formed 3 coherent clusters with a Silhouette Score of 0.9829 (scores near +1 indicate well-separated, compact clusters). | Model selection criterion: weighted F1 (classification) or RMSE (regression). All models trained with default hyperparameters; recommend GridSearchCV / Optuna for production hyperparameter optimisation.

[illegible]

Feature importance scores are not directly extractable for this model type. Consider using SHAP for model-agnostic explanations. | Importances sourced from estimator's native `feature_importances_` (tree-based) or `mean(|coef|)` (linear). For model-agnostic explanations, integrate SHAP `TreeExplainer` / `KernelExplainer`.

Throughput Speed : 48.5 MB/s (40,785 records)

PySpark RAM Req. : 512.0 MB (2 executors)

Partition Column : 'pay_period' (4 partitions)

Storage Format : Delta Lake / Apache Parquet (ACID compliant)

Compliance Rating: 75 / 100 (ISO 27001 Aligned)

EU AI Act Risk : Minimal Risk (Unsupervised Data Exploratory Analytics)

GDPR Privacy : Clean – No raw PII stored

Human Oversight : Mandatory (Human operator must approve critical fault actions)

→ Voluntary code of conduct and standard documentation.

1. Address missing data in columns ['max_salary', 'med_salary'] using automated sensor telemetry interpolation.

2. Investigate extreme sensor spikes detected in columns ['job_id', 'max_salary'] via IQR bounds.

3. Evaluate cluster cohesion (Silhouette Score: 0.9829) using PCA 2D projections.

4. Partition Delta Lake tables on column 'pay_period' for sub-second distributed telemetry queries.

5. Deploy the saved model binary via the FastAPI /orchestrate endpoint for production inference.

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End of ADSLM Report | ABB Evaluation Submission
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